Scope

The scope of this initiative is to deliver a ready-to-use Python analysis environment for the Technology Intelligence Platform (TIP) tailor-made for PATLIB centres. The aim is to ensure that TIP meets the most common and essential needs of PATLIB centres, particularly those whose staff lack the knowledge or time to learn Python in depth.
The solution should therefore make TIP easier to use, more accessible, and more relevant for the PATLIB network, enabling even entry-level users to explore and analyse patent and innovation data effectively without relying on programming skills.

The results of this initiative will form the basis of a pilot scheme starting in 2026.

Type of contribution

The contribution expected from the expert is the design and delivery of a ready-to-use Python analysis environment, based on the feedback collected from PATLIB interviews.
The overall goal is to deliver a solution that allows quick and flexible analysis, with particular focus on:

- Geographical and regional analysis: comparison of patent activity across country, regions, or specific areas.
- Sectoral or technological analysis: exploring trends across key technology domains.
- Multi-variable analysis, giving the user the possibility to combine external socioeconomic indicators to enable the exploration of possible correlations.

Specifically, the expert should develop (with the help and collaboration of the PATLIB team):

- A set of customisable queries that can be easily adapted by non-technical users (e.g., by simply changing parameters such as region, CPC/IPC class, or time period).
- An intuitive interface or script structure where users can generate meaningful visualisations and insights without modifying the core code.

Prep work

To identify the most relevant use cases and technical priorities, six structured interviews were conducted with PATLIB staff from different centres.

The majority of respondents were entry-level TIP users, often with limited time to invest in the tool, but who clearly recognised the potential value of TIP for improving their analytical and advisory activities.

From these discussions, several recurring needs and expectations emerged, including:

1. **Customisation and usability**

- Strong need for a customisable and intuitive interface that allows users to easily explore and compare patent data without requiring advanced coding skills.
- Users expect flexible dashboards and visualisations, where they can adjust parameters and filter dynamically to analyse trends by applicant, region, or technology.
- The ideal environment would include pre-set templates and query structures, enabling users to focus on analysis rather than technical setup.
- Simplified search functionalities (e.g., more accurate keyword-based searches) and clear visual outputs are seen as crucial for usability and adoption.

2. **Comparative and regional analysis**

- High demand for tools enabling comparative territorial analysis, such as identifying differences in innovation activity across NUTS regions or states.
- Interest in sector-specific views, allowing users to monitor patent dynamics in key fields.

- Preference for a system that supports cross-regional benchmarking, for example, comparing companies or applicants between different regions.
- Request for time-based analysis, such as tracking patent applications per CPC class over 5, 10, 15 years, and identifying emerging hotspots or declining fields.

3. Advanced analysis and data interpretation

- Users emphasised the importance of going beyond descriptive statistics towards analytical comparisons and data correlations (e.g., between patent volumes).
- The tool should provide light but insightful analytics, helping users identify key developments and make evidence-based recommendations for innovation policy or regional strategy.
- Ensuring data quality and interpretability is essential, as users want to understand what can realistically be achieved with the available data and visual outputs (example of data merging with EUROSTAT, World Bank Group, OECD sources could be an important output).

Expected outcome

By the end of the initiative, the goal is to have a functional prototype that:

- demonstrates several readymade analytical use cases tailored to the real needs of PATLIBs-
- enables non-technical users to perform regional, sectoral, and comparative analysis independently.
- provides a foundation for further development, potentially to be scaled or shared across the PATLIB network later on.
- features an interface that allows users to easily explore and compare patent data without requiring advanced coding skills.